

# Hojun Yang

M.S. Researcher - Robotics Field Autonomy & Geospatial Data Engineering

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## RESEARCH INTERESTS

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Field robotics and automation for mining and hazardous environments: GPS-denied underground exploration and navigation; mobile manipulation (robot arm) and robot learning for mining-equipment automation; UAV-based spatial AI and photogrammetry; and planetary surface robotics and in-situ resource utilization (ISRU).

## EDUCATION

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### M.S. in Energy Resources and Geosystems Engineering

Mar 2025 – Expected Feb 2027

Sejong University, Seoul, South Korea · SMART-X Lab · Advisor: Prof. Yosoon Choi

Thesis: Developing an AI-based UAV photogrammetry pipeline that removes mining equipment from image datasets and improves Digital Elevation Model (DEM) reconstruction accuracy in open-pit mines

Coursework: Microwave Remote Sensing, Geostatistics, Electro-Chemical Engineering, Energy Economics

GPA: 4.30 / 4.5

### B.S. in Energy Resources Engineering

Mar 2020 – Feb 2025

Pukyong National University, Busan, South Korea · Advisor: Prof. Yosoon Choi

Thesis: Autonomous Navigation and Automated Logistics System Using Smart Robots

Coursework: Machine Learning, Robotics, Mineral Processing, Geology, Rock Mechanics, Geothermal, Smart Grid

GPA: 4.34 / 4.5 (Class rank: 1 of 13)

## JOURNAL PUBLICATIONS

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**Yang, H.**, Ahn, T., & Choi, Y. (2026). Development of a YOLO-based detection model for key visual defect elements of tower cranes using UAV images. *Journal of the Korean Society of Mineral and Energy Resources Engineers*.

**Yang, H.**, & Choi, Y. (2026). Generation of vehicle-free digital elevation models in open-pit mines using UAV image inpainting. *Measurement*, 260, 119902. [SCIE]

**Yang, H.**, Kim, S., & Choi, Y. (2025). Development of a drone- and AI-based framework for exterior inspection and damage classification of tower cranes. *Tunnel and Underground Space*.

**Yang, H.**, Kim, S., & Choi, Y. (2025). Review of UAV–AI-based automated visual inspection technologies and their applicability to high-risk construction machinery. *Journal of the Korean Society of Mineral and Energy Resources Engineers*.

Jang, M., **Yang, H.**, & Choi, Y. (2025). Development of augmented reality-based educational content for mine workers. *Journal of the Korean Society of Mineral and Energy Resources Engineers*.

Choi, S., **Yang, H.**, Kim, Y., & Choi, Y. (2025). Router-drop-based robot teleoperation system for unconnected underground mines. *Journal of the Korean Society of Mineral and Energy Resources Engineers*.

**Yang, H.**, & Choi, Y. (2025). Analysis of wheel slip and path errors based on steering mechanisms of a small autonomous robot in underground rough environments. *Tunnel and Underground Space*.

## CONFERENCE PRESENTATIONS

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**Yang, H.**, & Choi, Y. UAV image inpainting for vehicle-free DEM generation in open-pit mines. *SME Annual Conference & Expo (MINEXCHANGE 2026)*, Salt Lake City, UT, USA, Feb 2026. (Poster)

**Yang, H.**, et al. Scaled collaborative-manipulator platform for robot-learning-based autonomous mining equipment. *KSMER Spring Conference*, Gyeongju, Korea, 2026. (Poster)

**Yang, H., & Choi, Y.** UAV image inpainting for open-pit mine DEM reconstruction. *KSMER Fall Conference*, Busan, Korea, 2025. (Oral) - **Best Paper Presentation Award**

**Yang, H., & Choi, Y.** Removing heavy equipment from open-pit mine DEMs via deep inpainting. *KSMER Spring Conference*, Jeju, Korea, 2025. (Oral)

**Yang, H., & Choi, Y.** Wheel slip and path error analysis of steering mechanisms in underground rough terrain. *KSMER Fall Joint Conference*, High1 Resort, Gangwon, Korea, 2024. (Oral)

## AWARDS & HONORS

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|-----------------------------------------------------------------------------|-----------------|
| Research Assistantship, Sejong University                                   | 2025-2026       |
| Future Talent Scholarship, Pukyong National University                      | 2025            |
| Academic Excellence Scholarship, Pukyong National University                | 2020, 2023-2024 |
| Best Paper Presentation Award, KSMER Fall Conference                        | 2025            |
| Gold Prize, Mine-Tech Festa (KOMIR) - GeoInpainter                          | 2025            |
| President's Award, Korean Society of Mineral and Energy Resources Engineers | 2025            |
| Commendation, Korean Association of Geographic Information Studies          | 2025            |
| Grand Prize, Mine-Tech Festa (KOMIR) - MeshRover Connect                    | 2024            |
| Excellence Award, Busan Future Scientist Award (Busan Sci. & Tech. Council) | 2024            |
| Excellence Award, Geoscience Data Utilization Competition (KIGAM)           | 2023            |
| Excellence Award, Mine-Tech Festa (KOMIR) - AR safety content               | 2023            |

## RESEARCH EXPERIENCE

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- Graduate Researcher - SMART-X Lab, Sejong University** 2026-2027
- **GeoInpainter - Vehicle-free Digital Elevation Model (DEM) generation:** built an automated pipeline (YOLOv8 detection → SAM masking → LaMa/MAT/DeepFillv2 inpainting → SfM) that removes heavy equipment from UAV imagery before 3D reconstruction, cutting elevation error from MAE 1.276 m to 0.111 m for a loader. First-author SCIE paper (*Measurement*, 2026).
  - **Mobile manipulation via vision-language-action models:** trained a UGV + 6-DoF collaborative arm for autonomous pick-and-place; forward-kinematics-based dataset auditing raised success to 90% (95-episode dataset). Current M.S. thesis direction.
  - **MeshRover Connect - Underground teleoperation:** co-developed a quadruped that autonomously drops mesh-network relay nodes to extend teleoperation in communication-denied mines; demonstrated in an active limestone mine. Grand Prize, Mine-Tech Festa 2024.
  - **Steering & wheel-slip analysis:** compared Ackermann, differential, and tracked steering on rough mine terrain using laser-grid ground truth plus simulation (best path RMSE 0.2311 m). First-author paper.
  - **Drone + AI inspection of high-risk machinery:** three-stage parallel YOLO architecture (ROI → part segmentation → defect detection) for UAV-based tower-crane inspection (defect mAP 0.937); South Korea government-funded.

## RESEARCH PROJECT

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**Development of a Robot Operating System (ROS) Extension Module for 3D Autonomous Navigation in Underground Rough-Terrain Environments (nuclear disaster site, lunar lava cave, abandoned mines)** 2023-2027

- Developed a router-drop-based robot teleoperation system for unconnected underground mines
- Utilized a thermal imaging camera to locate victims and sends a signal upon detection
- The system allows one potentially to investigate abandoned mines, nuclear disaster and lunar lava caves

**Development of Safety Assessment and Management Technology for High-Risk Construction Machinery** 2025-

- Proposed a systematic framework with ten major structural components of tower failures and seven major categories for defining inspection items and classifying damage types

- Established quantitative criteria for each category to standardize AI training datasets and support the development of automated damage detection models
- The proposed framework provides a technical foundation for constructing smart inspection platforms and advancing automation technologies for the safety assessment of construction equipment

### **Development of a Compact Planetary Rover System for the Korea Space Rover Challenge (KSRC) 2026**

2026-Organized by the Korea AeroSpace Administration (KASA)

- Designed an automated planetary-surface robot engineered for stable locomotion across mixed regolith conditions, including salt-rich and sandstone terrains.

### **TECHNICAL SKILLS**

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**Robotics & autonomy:** ROS (Noetic); LiDAR SLAM (FAST-LIO/FAST-LIO2, LeGO-LOAM); elevation mapping & traversability analysis; MoveIt; behavior trees; Gazebo simulation

**Perception & deep learning:** YOLOv8/v11; SAM; image inpainting (LaMa, MAT, DeepFillv2); OpenCV; PyTorch

**Robot learning:** imitation learning; ACT; SmolVLA / LeRobot

**Photogrammetry & GIS:** UAV mapping; Structure-from-Motion (Pix4D, Metashape); DEM & point-cloud analysis; GeoAI; QGIS

**Hardware & platforms:** AgileX Scout Mini / LIMO; Unitree Go1; MyCobot Pro 600; Velodyne VLP-16; Intel RealSense D435i/D455; Xsens IMU; NVIDIA Jetson AGX Orin

**Programming:** Python; Linux/shell

### **LANGUAGES**

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Korean (native); English (research & professional working proficiency)